

Kshitij Gupta

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Open to remote worldwide

SUMMARY

ML Engineer delivering production systems across computer vision, generative AI, and cloud-native MLOps. Shipped end-to-end pipelines from sub-5 MB on-device TFLite models to Kubernetes inference clusters on GCP & Azure serving Fortune-500 clients.

EDUCATION

IIIT Hyderabad Aug 2021 – May 2023
M.Tech, Computer Science — *NLP, Statistical AI, Distributed Systems, Blockchains, Neural & Cognitive Modelling*
IIIT Delhi Aug 2015 – May 2019
B.Tech, Computer Science — *Computer Vision, Machine Learning, GPU Computing, Image Analysis*

SKILLS

Languages Python, C, C++, JavaScript, TypeScript
AI / ML PyTorch, OpenCV, Scikit-Learn, ONNX, TFLite, Triton Inference Server
LLM / RAG LangChain, Qdrant, RAGAS, LLM-as-judge, Sentence Transformers, Groq, OpenAI, Anthropic
Web Stack FastAPI, Django, Flask, Node.js, React.js, Next.js, Electron.js
Cloud / MLOps Docker, Kubernetes, KServe, GCP, Azure, Proxmox, GitHub Actions, CI/CD | **DB:** PostgreSQL, ChromaDB

EXPERIENCE

CARS24 — *Machine Learning Engineer* Gurgaon | June 2024 – June 2025

- Designed and deployed end-to-end **Image Validation Pipeline** for per-part car inspection via **KServe + Triton** on GCP Kubernetes, cutting inspection time by **20–30 min per vehicle** and eliminating manual review bottlenecks.
- Engineered on-device **TFLite** model for part classification & pose detection — full binary **under 5 MB**, enabling offline inference on Android devices with zero cloud round-trip latency.
- Extended damage-estimation pipeline with **underbody oil-leak detection** module, improving inspection coverage for a previously undetected damage category.
- Owned the complete **MLOps lifecycle**: CVAT annotation → dataset versioning → training → automated cloud deployment, eliminating manual handoffs across teams.

Typeface India — *Software Engineer* Hyderabad | July 2023 – March 2024

- Shipped production generative-image features for Fortune-500 clients (**LG, Marks & Spencer, Ford**), owning the full pipeline from model integration to deployed inference.
- Integrated and productionised **Stable Diffusion, Google ImageGen & DALL-E**; built pipelines for scene generation, inpainting, outpainting, colour-palette extraction, and style-transfer at enterprise scale.
- Maintained high-availability inference infrastructure on **Azure & GCP** for masking-based and generative workloads, ensuring SLA compliance for Fortune-500 clients.

ttddsoft — *Software Engineer (Founding)* New Delhi | May 2020 – June 2021

- Built cross-platform **DRM suite (Windows, macOS, Android)** — ePendrive, Secure Browser, encrypted video player with screen-capture protection; re-architected core stack from Python to **Electron.js** for performance and reliability.
- Developed **Node.js** licensing & encryption API; self-hosted entire stack (DBs, APIs, email OTP) on bare-metal with zero cloud spend.

MirrorSize US Inc. — *AI Engineer* New Delhi | May 2019 – June 2020

- Built **Mask R-CNN** person-segmentation pipeline for automated body-contour extraction and body-measurement estimation from single-shot images, enabling virtual fit without specialist hardware.

PROJECTS

LLM Eval Harness [Link](#)

Automated evaluation pipeline for RAG & agentic LLM systems | Python, FastAPI, ChromaDB, PostgreSQL

- Built a YAML-driven evaluation harness supporting **4 LLM providers** (Ollama, Groq, OpenAI, Claude) with **LLM-as-judge scoring** for faithfulness, relevance, hallucination rate, and latency metrics.
- Implemented a full **RAG evaluation pipeline** using ChromaDB and sentence-transformers for document ingestion, embedding, retrieval, and automated benchmarking across multiple models.
- Developed a **FastAPI dashboard**, HTML reporting system, and **GitHub Actions CI/CD gate** that blocks deployments when hallucination thresholds regress beyond configured limits.

Image Similarity Search [Link](#)

CLIP + FAISS based semantic image retrieval engine | Python, FastAPI, PyTorch, FAISS

- Built a production-style image similarity search system using **CLIP ViT-B-32** embeddings and **FAISS ANN indexing** to retrieve semantically similar images with sub-millisecond search latency.
- Implemented **HNSW**, IVF, and exact FAISS indexes with benchmarking for recall vs latency trade-offs; added incremental indexing, metadata filtering, and embedding caching.
- Developed a **FastAPI backend** and **Streamlit frontend** supporting image uploads, top-K retrieval, similarity scoring, metrics, and real-time semantic search visualisation.

NeuralDocs AI [Link](#)

Local-first AI document intelligence platform | FastAPI, Ollama, ChromaDB, Docker, Next.js

- Self-hosted PDF analyser running **LLM inference locally via Ollama** — zero data egress; deployable on VPS, Proxmox, or bare-metal via **docker compose** up. Delivers summaries, structured Q&A, and document insights.

ONNXLab [Link](#)

Browser-native ONNX visualization & inference platform | Next.js, TypeScript, React Flow, ONNX Runtime Web

- Built an interactive **ONNX model visualization platform** with dynamic graph rendering, tensor metadata inspection, and Netron-inspired node analysis using React Flow and Dagre.
- Implemented fully client-side **ONNX inference in the browser** using ONNX Runtime Web with automatic **WebGPU/WASM fallback** for GPU-accelerated execution.
- Developed a dynamic tensor input pipeline supporting image preprocessing, tensor analysis, top-K prediction extraction, and custom label mapping for classification models.

ACHIEVEMENTS

- Dean's List, IIIT-H (M.Tech) | **KVPY Scholar** 2014 (IISc) | **Google Lunar X-Prize Moonbots** Top-30 Finalist | **JSTSE Scholar** 2012 (Delhi Govt.)